CLINICAL · EAR

Otitis externa and skull base osteomyelitis

Most otitis externa is a painful, swollen canal that needs cleaning and the right drops, not oral antibiotics. The exception changes everything: an older diabetic with severe night pain and granulation in the canal has skull base osteomyelitis until proven otherwise.

AT A GLANCE

- Painful canal with tragal tenderness and a mobile drum.
- Clean it, wick it, use drops; oral antibiotics are rarely needed.
- Fungal disease itches and shows spores.
- Night pain and granulation in a diabetic is skull base osteomyelitis.
- Technetium confirms; gallium follows; treat about 6 weeks.
- Necrotising otitis externa: biopsy, MRI or PET-CT, dual antipseudomonal therapy for at least 6 weeks.

1 CHAPTER ONE Overview

Infection of the ear canal skin.

1.1 Definition

Inflammation of the skin of the external auditory canal, bacterial, fungal or reactive. The canal is a blind, warm, dark tube lined by thin skin over bone and cartilage, and anything that breaks its defences (moisture, scratching, loss of wax) lets infection in.

1.2 Forms

OTITIS EXTERNA

- BACTERIAL **Acute diffuse otitis externa** furuncle (localised)
- FUNGAL **Otomycosis** Aspergillus, Candida

- REACTIVE **Eczematous, seborrhoeic** contact dermatitis from drops or moulds, neurodermatitis
- CHRONIC **Chronic otitis externa** stenosis of the canal
- INVASIVE **Necrotising (malignant) otitis externa** skull base osteomyelitis

1.3 Applied anatomy

The outer third of the canal is cartilaginous, with hair follicles and sebaceous and ceruminous glands; the inner two-thirds is bony with thin, gland-free skin. Infection spreads from the canal through the *fissures of Santorini* in the cartilage and the tympanomastoid suture to the parotid, the skull base and the temporomandibular joint.

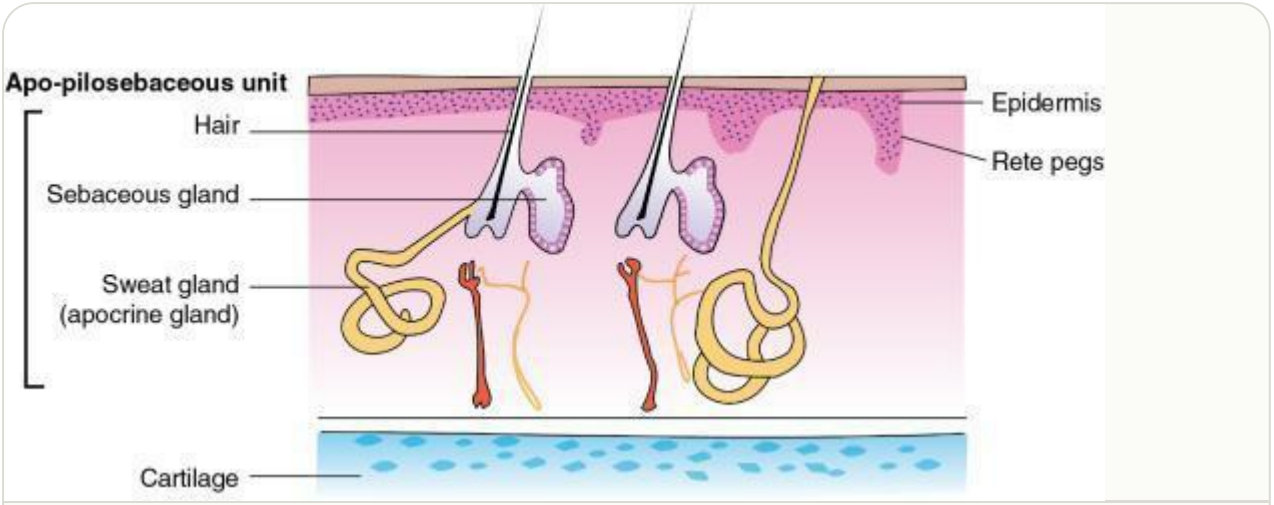


FIGURE 1 Skin of the cartilaginous canal, with its hair follicles and sebaceous and ceruminous glands.
Dhingra, fig. 8.12

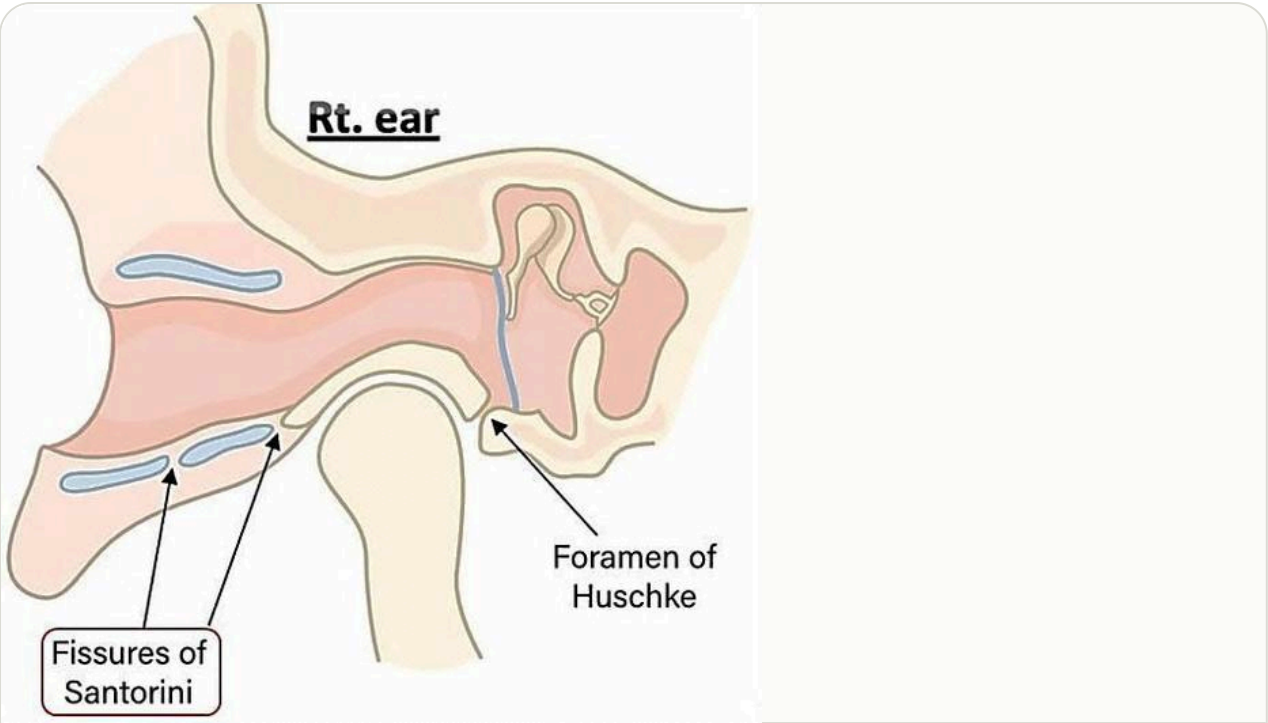


FIGURE 2 Fissures of Santorini and the foramen of Huschke, routes from the canal to the parotid and skull base.

Anatomy atlas · ENT Steps p. 32

CHAPTER TWO

2 Acute diffuse otitis externa

Swimmer's ear.

2.1 Causes

Moisture (swimming, humid climates), trauma from cotton buds and fingernails, hearing aids and earbuds, eczema and loss of protective wax. *Pseudomonas aeruginosa* and *Staphylococcus aureus* are the usual organisms.

2.2 Clinical picture

2.2.1 Symptoms

Pain that develops over a day or two, itching, a feeling of blockage and a scanty discharge.

2.2.2 Signs

Pain on moving the tragus or pinna; a red, swollen canal with debris; tender nodes in front of or behind the ear. When visible, the drum moves normally on pneumatic otoscopy, unlike otitis media.

2.3 Management

2.3.1 Aural toilet

Clean the canal under vision by suction or dry mopping. Debris collects in the anteroinferior recess, so look there.

OPERATIVE GUIDES IN THEATRE 1 operation

Office-Based Procedures in Otology
Office debridement

2.3.2 Wick

If the canal is swollen shut, insert an expanding wick soaked in antibiotic-steroid drops, or in an astringent such as aluminium acetate 8% or silver nitrate 3%.

WHERE THE BOOKS DIFFER · CHANGING THE WICK

Dhingra
Daily for two to three days.

Lee
Every three to five days.

2.3.3 Drops

Antibiotic-steroid drops for a week to ten days. Acidifying with dilute vinegar helps mild cases and prevention.

2.3.4 Oral antibiotics

Not needed, unless there is spreading cellulitis or the patient is diabetic or immunocompromised.

2.3.5 Keep it dry

No swimming until healed; cotton wool with petroleum jelly when bathing.

WHEN DROPS FAIL

> Think of contact allergy to neomycin, fungal superinfection, or an underlying necrotising infection.

3 CHAPTER THREE Furuncle

A boil in the canal.

3.1 Picture

A staphylococcal infection of a hair follicle, so it occurs only in the hair-bearing outer cartilaginous canal. Pain is out of proportion to its size and worse on chewing; it can swell the postauricular groove and mimic mastoiditis.

3.2 Distinguishing from mastoiditis

The drum is normal, there is no preceding otitis media and no mucoid discharge, and pinna movement is painful.

3.3 Treatment

Analgesia, oral antistaphylococcal antibiotics, heat, and an ichthammol-glycerine pack; incise when it points. Recurrent furuncles call for testing for diabetes and treating nasal staphylococcal carriage.

4 CHAPTER FOUR Otomycosis

Itch and spores.

4.1 Organisms

Aspergillus niger (black-headed spores), *Aspergillus fumigatus* (pale blue or green) and *Candida* (creamy white). Repeated antibiotic drops, humidity and swimming

predispose.

4.2 Picture

Intense itching, discomfort, blockage and a watery discharge with a musty smell. The canal contains a mass like *wet newspaper*, with black, grey or white spores.

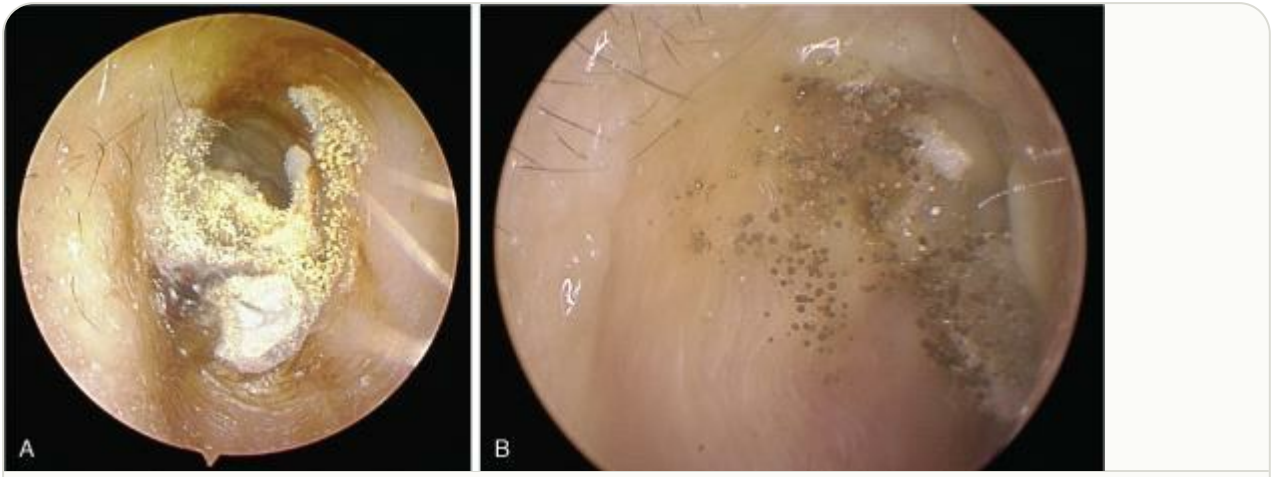


FIGURE 3 Endoscopic view of otomycosis with yellow (A) and black (B) fungal spores.
Dhingra, fig. 8.11

4.3 Treatment

Meticulous cleaning, then a topical antifungal, continued for a week beyond apparent cure: clotrimazole for most, nystatin for *Candida*. Salicylic acid in alcohol and povidone iodine are alternatives.

5 CHAPTER FIVE Chronic and reactive otitis externa

Itch, scaling and narrowing.

5.1 Chronic otitis externa

Long-standing itch and irritation with thick, dry, scaling skin, which may narrow the canal. Treat underlying eczema or seborrhoeic dermatitis, use steroid drops or

ointment, and stop the patient scratching.

5.2 Reactive forms

Eczematous otitis externa from drops or hearing-aid moulds, seborrhoeic dermatitis with scalp involvement, and neurodermatitis from habitual scratching.

5.3 Otitis externa haemorrhagica

Blood blisters on the deep canal and drum, often with influenza.

5.4 Chronic stenosing otitis externa

Fibrous narrowing of the deep canal. Treated by excising the thick skin, widening the bony canal and grafting.

6 CHAPTER SIX Necrotising otitis externa

The dangerous one.

6.1 What it is

An invasive *Pseudomonas* infection (about 90% of cases) that spreads from the canal to the bone of the skull base: a skull base osteomyelitis. It affects older diabetics and the immunocompromised. *Aspergillus* causes a minority.

6.2 Clinical picture

6.2.1 Symptoms

Severe, deep, unrelenting pain, worse at night and out of proportion to what is seen, with a persistent discharge despite treatment.

6.2.2 Signs

Granulation tissue on the floor of the canal at the bony-cartilaginous junction. Cranial nerve palsies mark spread: the facial nerve first at the stylomastoid

> foramen, then the ninth, tenth and eleventh at the jugular foramen, and the twelfth. Multiple palsies carry a worse prognosis.

6.3 Investigations

WHICH SCAN?

TEST	USE	LIMITATION
Technetium-99 bone scan	Confirms early bone involvement	Stays positive for months
Gallium or indium scan	Follows response; repeated every few weeks	Less anatomical detail
CT	Bone erosion	Changes appear late
MRI	Soft tissue, marrow and dura	Not a guide to cure

Add ESR or CRP (useful to follow), glucose, culture, and biopsy of the granulation to exclude carcinoma.

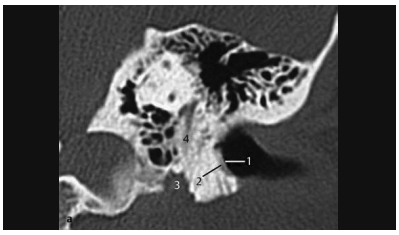
WHERE THE BOOKS DIFFER · MONITORING AND DURATION

Dhingra
Gallium repeated every three weeks; treatment for six to eight weeks, including intravenous aminoglycoside combinations.

Lee
Gallium every four weeks; oral fluoroquinolone for about six weeks as first line.

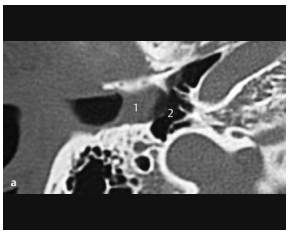
IMAGING IN THE APP 3 cases

Necrotising otitis externa, canal stenosis and canal carcinoma.



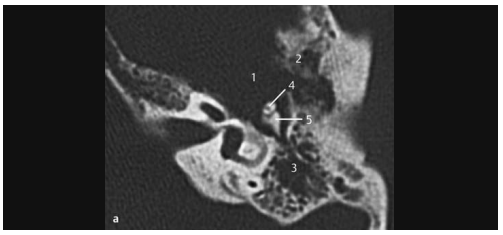
Malignant External Otitis or Necrotizing External Otitis

Tumours · 6 images



Stenosis of the External Auditory Canal

Bone disease · 2 images



Malignancy of the External Auditory Canal Squamous Cell Carcinoma

Tumours · 2 images

6.4 Treatment

Admit; control the diabetes; regular cleaning; high-dose oral ciprofloxacin, or intravenous anti-pseudomonal antibiotics for advanced disease; voriconazole for *Aspergillus*. Continue until the examination, inflammatory markers and scans have normalised. Surgery is limited to biopsy and removal of sequestra or abscesses. Hyperbaric oxygen is unproven.

6.5 Outcome

Mortality remains around 5 to 20%, from sepsis, sinus thrombosis and cranial neuropathies. Quinolone resistance is rising.

SUSPECT IT WHEN

- A diabetic or immunocompromised patient has severe pain and granulation in the canal.
- Otitis externa fails to respond after a week of proper treatment.
- A cranial nerve palsy appears with ear pain.

NECROTISING OTITIS EXTERNA: CURRENT PRACTICE (OXFORD TEXTBOOK)

Infection spreads from the canal floor through the **fissures of Santorini** into the infratemporal fossa and skull base. Between 65 and 100% of patients are diabetic, but the classic "elderly diabetic with a painful discharging ear" is not always present, especially in other forms of immunosuppression. *Pseudomonas aeruginosa* is the usual organism, about a quarter of isolates are resistant, and fungi are cultured in over 20%

of cases, more often in HIV with low CD4 counts and with a worse prognosis. Surface swabs may grow nothing, so send both a swab and a **granulation biopsy** for bacterial and fungal culture and for histology, because squamous carcinoma of the temporal bone is the key differential; if there is no tissue in the canal, deep samples may be taken at cortical mastoidectomy.

Imaging. CT shows erosion only after about 30% demineralisation, so a normal early CT is falsely reassuring. Contrast MRI with fat suppression and diffusion shows soft tissue and marrow disease earlier and detects intracranial spread. Gallium and technetium scans have largely been replaced by MRI, and **FDG PET-CT** is increasingly used to diagnose and to monitor response. CT and MRI changes persist for up to a year after successful treatment, so they cannot be used to decide when to stop.

Treatment. Admit for multidisciplinary care (otology, diabetes, microbiology, radiology, elderly care), start empirical intravenous antipseudomonal therapy at once, and give **dual therapy**, typically piperacillin–tazobactam or ceftazidime with a fluoroquinolone, adjusted to cultures. There is no agreed guideline, but consensus favours at least **six weeks** of intravenous treatment, often completed at home, guided by clinical response, inflammatory markers and imaging. Surgery is now limited to biopsy, deep sampling and drainage of abscesses; debridement and nerve decompression have no proven benefit.

CHAPTER SEVEN

7 For the examination

What the examiner wants to hear.

7.1 On call

A PAINFUL, SWOLLEN EAR CANAL

1. Clean under vision; insert a wick if the canal is closed.
2. Antibiotic-steroid drops; analgesia; keep dry.
3. Check glucose in diabetics and anyone with recurrent infection.
4. Diabetic or immunocompromised with severe pain and granulation: biopsy, ESR, imaging and admit.

7.2 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Tragal tenderness** and a mobile drum versus otitis media.
- **Furuncle** in the cartilaginous canal only.
- **Otomycosis:** organisms and appearance.
- **Necrotising otitis externa:** who, organism, signs, scans and treatment.

7.3 Common pitfalls

COMMON PITFALLS

- Prescribing oral antibiotics instead of cleaning and drops.
- Missing a necrotising infection in a diabetic.
- Neomycin contact allergy mistaken for treatment failure.
- Drops into a canal full of debris.

Written for this app from Oxford Textbook of Otolaryngology ch. 8, 13, Oxford Specialist Handbook ch. 15, Oxford Handbook of ENT ch. 5, Dhingra ch. 8 and K. J. Lee ch. 22. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

EXTERNAL EAR & TYMPANIC MEMBRANE · TUMOURS

Malignant External Otitis or Necrotizing External Otitis

IMAGING FINDINGS

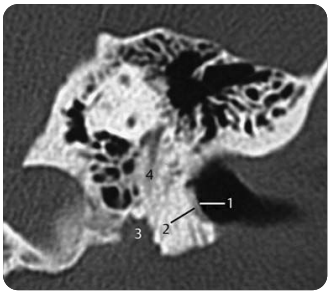
- Soft tissue in the external canal with erosion of its floor and the tympanic bone
- Spread below the skull base: loss of fat at the stylomastoid foramen, parapharyngeal and masticator spaces, temporomandibular joint
- Later: skull base osteomyelitis of the clivus, jugular foramen and petrous apex; venous thrombosis

TOP DIFFERENTIALS

- Carcinoma of the external canal
- Canal cholesteatoma
- Keratosis obturans

CHECKLIST

- Facial nerve and lower cranial nerves
- Sigmoid sinus and jugular vein; follow-up MRI changes lag behind clinical cure



CT, coronal. Necrotizing lesion in the floor of the external auditory canal (1). Clinically, erosive bone lesions (2) with purulent debris and denuded bone…



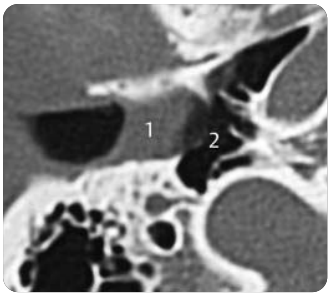
Axial bone CT shows EAC opacification with focal anterior wall and floor of middle ear erosion in this diabetic patient with painful otorrhea and earl…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1010.

EXTERNAL EAR & TYMPANIC MEMBRANE · BONE DISEASE

Stenosis of the External Auditory Canal

- Beware of accumulation of cerumen or epithelium behind the fibrous wall, and exclude deeper pathology in the middle ear. In most cases, the fibrous wall can be surgically separated from the inner layers of the eardrum without opening the middle ear.
- In cases of malignant external otitis or secondary cholesteatoma, deeper pathologies (e.g., to the facial nerve) are of special interest.

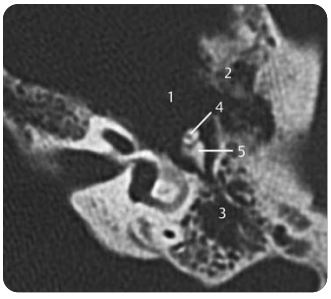


CT, axial. Chronic external otitis may lead to cicatrizing fibrous changes in the wall (1) of the external auditory canal. In this patient, the fibrous wall i…

EXTERNAL EAR & TYMPANIC MEMBRANE · TUMOURS

Malignancy of the External Auditory Canal Squamous Cell Carcinoma

- Extension of bony destruction and features suggestive of infiltrative growth are associated with high morbidity, such as dysfunction of the facial nerve and inner ear.
- Adherence to vital structures, such as the internal carotid artery, may preclude surgery.
- Secondary complications, i.e., mastoiditis or otitis media, as a result of stasis of secretions due to obstruction of the eustachian tube. After radiotherapy, necrotizing osteomyelitis and mastoiditis may occur.



CT, axial. Cranial slice at the level of the horizontal canal, showing destruction of the wall of anterior epitympanum (1), irregular invasion of…